Does Role Specialization Matter for Explanation Faithfulness in Mixture-of-Experts? Supplementary Material

Yeji Kim[1], Housam Babiker[2], Mi-Young Kim[3], and Randy Goebel[1]

[1] University of Alberta, Edmonton, AB, Canada
{yeji7,rgoebel}@ualberta.ca
[2] University of Waikato, Hamilton, New Zealand
housam.babiker@waikato.ac.nz
[3] University of Alberta, Camrose, AB, Canada
miyoung2@ualberta.ca

1 Experimental Details

1.1 Hyperparameter Configurations

All hyperparameter settings are from I^{2}MoE [5]. Table 1 reports the final hyperparameter configurations used for MIMIC, MMIMDb, and ENRICO. Modality abbreviations follow the notation in the main paper.

Table 1. Final hyperparameter configurations used for the experiments on MIMIC, MMIMDb, and ENRICO.

Hyperparameter	MIMIC	MMIMDb	ENRICO
Learning rate (`lr`)	0.0001	0.0001	0.0001
Temperature for reweighting (`temperature_rw`)	1.0	2.0	2.0
Hidden dimension for reweighting (`hidden_dim_rw`)	256	256	256
Number of layers in reweighting (`num_layer_rw`)	3	2	3
Interaction loss weight (`interaction_loss_weight`)	0.01	0.5	0.5
Modality	LNC	LI	SW
Training epochs (`train_epochs`)	30	40	50
Batch size (`batch_size`)	32	32	32
Number of encoder layers (`num_layers_enc`)	1	2	2
Number of fusion layers (`num_layers_fus`)	2	2	2
Number of prediction layers (`num_layers_pred`)	2	2	2
Number of attention heads (`num_heads`)	4	4	4
Hidden dimension (`hidden_dim`)	128	256	256

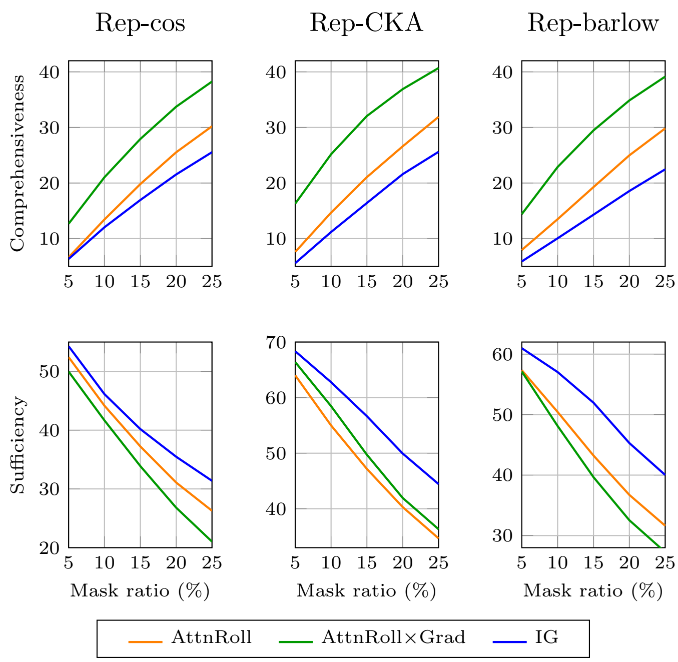

Fig. 1. Comparison of attribution methods on ENRICO. The top row reports comprehensiveness and the bottom row reports sufficiency across different mask ratios. AttnRoll×Grad shows the strongest and most consistent faithfulness trends across Rep-cos, Rep-CKA, and Rep-barlow, and is therefore used in the main experiments.

2 Attribution Method Selection

2.1 Comparison on ENRICO

Before fixing the attribution method used in the main paper, we compared three alternatives on ENRICO: Attention Rollout (AttnRoll) [1], Attention Rollout×Gradients (AttnRoll×Grad) [2], and Integrated Gradients (IG) [4]. We evaluate them with comprehensiveness and sufficiency under three representation variants: cosine similarity, CKA and Barlow-style similarity. Higher comprehensiveness and lower sufficiency indicate better faithfulness.

Figure 1 shows the full curves across perturbation budgets from 5% to 25%. Across all three representation metrics, AttnRoll×Grad consistently achieves the highest comprehensiveness. For sufficiency, it is either the best or a close second depending on the representation metric, yielding the most stable overall trade-off among the compared methods. Based on this observation, we use AttnRoll×Grad in all subsequent experiments in the main paper.

Table 2. Raw comprehensiveness under identified and random masking. Higher identified comprehensiveness indicates better faithfulness; random values are reported as a control for gap computation.

Dataset	Mask	Baseline		Rep-Cos		Rep-CKA		Rep-Barlow	
		Identified	Random	Identified	Random	Identified	Random	Identified	Random
ENRICO	5%	6.90±1.85	1.32±0.33	12.76±2.82	1.17±0.24	18.70±8.96	1.42±0.23	13.44±5.85	1.03±0.12
	10%	13.27±2.93	3.19±0.69	21.05±4.70	2.70±0.56	29.10±11.69	3.43±0.42	21.97±8.08	2.66±0.05
	15%	19.30±3.67	5.42±0.86	28.38±5.68	4.57±0.91	36.87±12.97	5.84±0.51	29.00±9.71	4.83±0.45
	20%	25.28±3.94	8.12±1.13	34.71±6.33	6.78±1.29	43.49±13.21	8.72±0.66	35.19±10.78	7.41±0.81
	25%	30.32±4.03	10.97±1.37	39.54±6.57	9.26±1.62	48.10±12.81	11.59±0.91	39.81±11.08	10.09±1.29
MIMIC	5%	2.92±0.42	0.43±0.03	5.73±3.87	0.41±0.05	6.88±1.53	0.35±0.04	6.73±1.54	0.34±0.04
	10%	3.82±0.94	0.86±0.07	6.55±3.80	0.87±0.11	8.77±2.39	0.74±0.13	6.27±1.16	0.80±0.14
	15%	4.73±1.33	1.29±0.09	6.53±3.69	1.29±0.19	10.43±3.48	1.11±0.21	6.20±0.95	1.15±0.15
	20%	5.51±2.15	1.79±0.16	7.50±3.75	1.79±0.33	11.42±3.74	1.51±0.28	6.37±1.09	1.61±0.24
	25%	6.26±2.54	2.31±0.18	8.15±4.36	2.33±0.38	12.36±3.41	2.01±0.36	6.80±1.31	2.06±0.29
MMIMDb	5%	12.11±1.38	0.62±0.04	17.03±3.78	0.75±0.10	15.19±0.77	0.65±0.14	15.20±2.32	0.76±0.15
	10%	18.58±2.13	1.34±0.05	24.99±4.89	1.48±0.20	22.67±1.21	1.41±0.28	22.94±3.12	1.61±0.34
	15%	23.72±2.67	2.20±0.10	30.99±5.50	2.34±0.28	28.39±1.47	2.32±0.40	28.90±3.53	2.62±0.54
	20%	27.82±3.09	3.16±0.17	35.45±5.80	3.28±0.34	32.90±1.68	3.33±0.56	33.52±3.81	3.75±0.75
	25%	31.24±3.34	4.28±0.22	38.96±5.95	4.29±0.41	36.49±1.79	4.42±0.68	37.13±3.90	5.01±0.98

3 Additional Faithfulness Results

3.1 Full Faithfulness Results

Table 2 and 3 report the full faithfulness results under both random and identified perturbations, including mean and standard deviation across three random seeds. These results complement the summarized presentation in the main paper by quantifying run-to-run variability and by separating generic performance degradation from degradation caused by removing features selected by the explanation method. The AOPC values [3] reported in the main paper correspond to the average over the five mask ratios shown here.

In the random setting, perturbations are applied to randomly chosen features and primarily serve as a control. In the identified setting, perturbations are applied to features highlighted by the attribution method, providing a more direct test of whether the explanations capture decision-relevant information. Higher comprehensiveness and lower sufficiency indicate better faithfulness in both settings.

3.2 General Sparse MoE

Following the main paper, we additionally report the omitted general sparse MoE results on MIMIC and MMIMDb. Table 4 summarizes the original predictive performance together with AOPC-based faithfulness scores averaged over three random seeds. For faithfulness, we report the same AOPC gap used in the main text: identified minus random for comprehensiveness, and random minus identified for sufficiency. The corresponding random/identified AOPC values are

Table 3. Raw sufficiency under identified and random masking. Lower identified sufficiency indicates better faithfulness; random values are reported as a control for gap computation.

Dataset	Mask	Baseline		Rep-Cos		Rep-CKA		Rep-Barlow	
		Identified	Random	Identified	Random	Identified	Random	Identified	Random
ENRICO	5%	56.33±1.49	59.39±2.29	47.77±3.96	62.58±5.08	64.94±5.02	72.17±7.90	55.71±7.58	64.43±4.46
	10%	51.34±0.38	57.33±1.96	38.71±4.44	58.85±4.47	55.44±6.73	68.31±7.17	44.98±6.96	61.66±4.96
	15%	45.93±2.15	54.89±1.41	30.91±3.43	55.06±3.77	46.48±8.75	63.70±5.92	35.82±6.24	58.18±4.82
	20%	40.46±3.21	52.08±1.33	23.78±1.64	51.02±2.76	38.15±9.91	58.75±4.23	28.91±6.27	53.84±3.53
	25%	35.28±3.45	49.45±1.81	18.45±0.69	47.25±1.99	31.96±9.74	54.90±3.38	23.85±6.09	49.90±2.51
MIMIC	5%	12.88±0.43	17.00±1.19	10.14±4.72	17.20±1.13	7.53±4.79	18.11±1.74	7.34±3.50	14.40±0.69
	10%	11.63±0.97	15.32±1.01	9.76±3.75	15.54±1.31	6.78±4.25	15.82±1.66	7.83±2.99	13.17±0.93
	15%	9.83±1.23	13.99±0.87	9.16±4.26	14.17±1.38	5.51±4.75	13.99±1.68	7.55±2.79	12.14±1.11
	20%	8.99±2.07	12.50±0.73	7.67±4.71	12.63±1.42	4.70±4.35	12.10±1.63	7.23±2.74	10.84±1.21
	25%	7.89±1.83	11.29±0.63	6.81±5.18	11.33±1.47	3.94±3.59	10.64±1.48	6.83±2.71	9.85±1.28
MMIMDb	5%	13.82±2.34	43.28±2.08	6.79±2.29	39.83±1.01	10.78±1.49	44.27±1.00	13.34±1.31	43.80±1.63
	10%	9.52±2.45	36.14±1.31	3.97±1.87	33.10±0.93	6.89±1.00	36.49±0.13	8.89±1.05	37.44±2.08
	15%	7.19±2.26	30.77±0.79	2.69±1.49	28.28±1.02	4.94±0.80	30.89±0.51	6.47±1.07	32.55±2.00
	20%	5.62±2.07	26.60±0.53	1.88±1.19	24.59±1.02	3.71±0.71	26.67±0.82	4.87±1.03	28.58±1.91
	25%	4.45±1.87	23.24±0.49	1.28±0.97	21.60±1.02	2.82±0.59	23.31±0.99	3.70±0.98	25.20±1.79

Table 4. General Sparse MoE results on MIMIC and MMIMDb. We report predictive performance and faithfulness AOPC gaps. The corresponding random/identified values are shown underneath each gap.

MIMIC

	Metric	Baseline	Rep-CKA
Task	Acc ↑	83.15±0.22	84.65±0.22
	Macro-F1 ↑	74.81±0.20	80.71±0.22
Faith.	AOPC Comp Gap ↑	15.88	50.74
	(rnd/id)	(0.91 / 16.79)	(0.88 / 51.62)
	AOPC Suff Gap ↑	12.17	20.36
	(rnd/id)	(8.13 / -4.04)	(13.87 / -6.49)

MMIMDb

	Metric	Baseline	Rep-CKA
Task	Micro-F1 ↑	67.03±0.11	67.21±0.11
	Macro-F1 ↑	56.53±0.11	56.70±0.10
Faith.	AOPC Comp Gap ↑	27.70	28.54
	(rnd/id)	(2.24 / 29.94)	(1.96 / 30.51)
	AOPC Suff Gap ↑	27.05	28.08
	(rnd/id)	(30.97 / 3.92)	(30.68 / 2.60)

shown underneath each gap. Higher gap values therefore indicate stronger separation between explanation-guided and random perturbations. Note that some sufficiency AOPC values on MIMIC are negative. This can occur when retaining only the attributed features removes counter-evidence and increases the target logit relative to the original input, so the sufficiency score remains well-defined and should not be interpreted as an implementation error.

On MIMIC, Rep-CKA improves both predictive performance and faithfulness relative to the baseline. On MMIMDb, Rep-CKA yields slightly better Micro-F1 and Macro-F1 while also modestly improving both faithfulness gaps.

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